ind

A multi-calendar summary for Emacs

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1 Introduction

`ind` is an Emacs Lisp successor to the Bash script of the same name. It presents a compact summary of the current date using numerous systems of year numbering, calendrical computation, traditional observance, and chronological reckoning.

The display may include the Byzantine year and indiction, Roman AUC, the Masoretic year, Japanese imperial dating, traditional Asian calendars, computistical values, astronomical day counts, Buddhist observances, quarter days, and several alternative calendars.

The command is intended as a daily calendrical reference rather than as a general-purpose date-conversion library.

2 Using ind

Invoke the display with:

```
M-x ind
```

The principal command has the following purpose:

Display a multi-calendar summary for today.

Show civil, historical, religious, astronomical, regnal, and idiosyncratic representations of the current date.

The individual values are described in the following chapters.

3 Dating Systems

3.1 Hebrew Calendar

The Hebrew or Jewish calendar is a lunisolar calendar used principally for Jewish religious observance and, together with the Gregorian calendar, for civil purposes in Israel.

Twelve lunar months are approximately eleven days shorter than a solar year. The Hebrew calendar therefore employs a 19-year Metonic cycle, inserting an additional month in seven years of each cycle to keep the months aligned with the seasons.

Hebrew years are numbered according to an *Anno Mundi*, or “year of the world,” era. The conventional epoch falls in 3761 BCE when expressed in the proleptic Julian calendar. Because Hebrew days begin at sunset and because several epoch conventions appear in secondary literature, apparently differing civil dates may refer to the same traditional epoch.

3.2 Roman AUC

Ab urbe condita, commonly abbreviated AUC, means “from the founding of the City.” The City is Rome, whose traditional foundation is assigned to 753 BCE in the chronology associated with Varro.

AUC numbering is convenient for modern chronological comparison, but it was not the ordinary Roman method of naming years. Romans commonly identified a year by the names of its consuls. Regnal years and other official eras were also used in particular periods and jurisdictions.

3.3 Byzantine Calendar

The Byzantine calendar is also called the Creation Era of Constantinople, the Byzantine Era, or the Byzantine *Anno Mundi*. It was used by the Byzantine Empire and by parts of the Eastern Orthodox world.

The system is based on the Julian calendar, but numbers its years from a calculated creation of the world. Its year begins on September 1. Year 1 extended from September 1, 5509 BCE, through August 31, 5508 BCE.

Consequently, the Byzantine year corresponding to a Gregorian date depends not only on the civil year but also on whether the date falls before or after September 1.

3.4 Indictions

An *indiction* is a year within a recurring 15-year cycle used in late Roman and medieval administration and document dating.

Each year of the cycle is numbered from 1 through 15. Since the cycles themselves were not normally numbered, an indiction number alone does not uniquely identify a historical year.

Different historical conventions began the indiction year on different dates. A calendrical program should therefore state or consistently apply the convention it uses.

3.5 Julian Period

The *Julian Period* is a chronological cycle of 7,980 years devised by Joseph Justus Scaliger. It combines three traditional cycles:

- the 15-year indiction cycle;
- the 28-year solar cycle; and
- the 19-year lunar or Metonic cycle.

The first year of the present Julian Period is 4713 BCE in conventional historical notation, corresponding to astronomical year −4712. The next Julian Period begins in 3268 CE.

The Julian Period should not be confused with the Julian calendar or with the Julian Date continuous day count.

3.6 Japanese Imperial Dating

Japanese dates may be expressed by imperial era name and regnal year. Each era begins with the accession or formal inauguration of a new emperor, and the first partial calendar year is numbered year 1.

A distinct nationalist year count, sometimes called the imperial year or *koki*, numbers years from the legendary foundation of Japan by Emperor Jimmu in 660 BCE. It is historically and conceptually different from ordinary era-name dating.

3.7 Chinese Calendar

The traditional Chinese calendar is lunisolar. Its months are related to lunar phases, while intercalary months keep the calendar aligned with the solar year and seasonal cycle.

Although the Gregorian calendar is used for ordinary civil dating, the traditional calendar continues to govern festivals such as Chinese New Year and the Lantern Festival. Traditional year designations may also combine a heavenly stem and earthly branch in a repeating 60-year cycle.

3.8 Tibetan Calendar

The Tibetan calendar is lunisolar and normally contains twelve lunar months. An intercalary month is inserted when required to keep the calendar aligned with the solar year.

The Phukpa tradition is the principal calendrical system used in much of Tibetan culture. Tibetan New Year, or *Losar*, ordinarily falls in February or March of the Gregorian calendar, though its date need not coincide with Chinese New Year.

3.9 Buddhist Era

The term *Buddhist calendar* refers to a family of related lunisolar and solar calendars used in mainland Southeast Asia and Sri Lanka for religious, traditional, or civil purposes.

The Buddhist Era is also a system of year numbering. Its offset from the Common Era differs according to regional convention and whether a date falls before or after the traditional new year. A displayed Buddhist year should therefore be understood in the context of the specific convention used by the program.

3.10 Masoretic Dating

In `ind`, *Masoretic dating* denotes a simple ceremonial or commemorative year count obtained by adding 4000 to the Gregorian year. It resembles an *Anno Mundi* system but should not be confused with the calculated year of the Hebrew calendar.

The name and offset are used here as a feature of the program's traditional output, not as a claim that this is a standard civil calendar.

4 Computistical Values

Computus is the body of calendrical calculation traditionally used to determine the date of Easter and related ecclesiastical dates.

4.1 Metonic Cycle and Golden Number

The Metonic cycle is a period of 19 solar years after which the phases of the Moon recur on approximately the same dates of the solar year.

The *golden number* assigns each year a number from 1 through 19, indicating its place in the Metonic cycle. Golden numbers were used in medieval computus and in perpetual and Runic calendars.

4.2 Epact

The *epact* expresses the age of the ecclesiastical Moon at the beginning of the solar year, or equivalently the accumulated difference between solar and lunar reckoning under a particular computistical system.

Because twelve lunar months are roughly eleven days shorter than the solar year, the epact normally advances by eleven from one year to the next, subject to the corrections of the calendar in use.

4.3 Dominical Letters

A *dominical letter*, or Sunday letter, is one of the letters A through G assigned to a year for finding Sundays in a fixed-date calendar.

A common year has one dominical letter. A leap year has two because the insertion of February 29 changes the correspondence between calendar dates and weekdays after February.

4.4 Ides

The *Ides* was one of the named reference days of the Roman month. It fell on the fifteenth day of March, May, July, and October, and on the thirteenth day of the other months.

Roman dates were ordinarily expressed by counting inclusively backward from the Kalends, Nones, or Ides rather than by numbering every day consecutively from the start of the month.

5 Continuous Day Counts

Continuous day counts identify dates or instants by numbering successive days without dividing them into calendar years and months.

5.1 Julian Date

The *Julian Date*, or JD, is a continuous count of days and fractions of a day from noon on January 1, 4713 BCE in the proleptic Julian calendar.

Because Julian Dates identify instants rather than merely civil dates, precise work should also state the time scale being used.

5.2 Modified Julian Date

The *Modified Julian Date*, or MJD, is defined by:

```
MJD = JD - 2400000.5
```

Its epoch is midnight at the beginning of November 17, 1858. The half-day subtraction moves the boundary from astronomical noon to civil midnight, while the larger subtraction shortens the displayed number.

5.3 Truncated Julian Date

The name *Truncated Julian Date*, or TJD, has been used for more than one shortened Julian-date convention.

The convention historically used by `ind` is associated with NASA's PB-5 spacecraft time code and counts days from MJD 44000, corresponding to May 24, 1968. The original field allowed a four-digit day count in a 14-bit representation; the later PB-5J form enlarged the field to 16 bits.

Because other technical sources use "truncated Julian date" for different subtractions, the epoch or formula should always accompany the abbreviation TJD.

5.4 Rata Die

Rata Die, abbreviated RD, is a continuous count of calendar days used in calendrical calculation. Day 1 is January 1, 1 CE in the proleptic Gregorian calendar.

The expression is Latin for "fixed day" or "reckoned day." Some implementations extend the integer day count with a fractional part for time of day.

6 Traditional Observances

6.1 Quarter and Cross-quarter Days

In British and Irish tradition, quarter days were customary dates for the beginning of school terms, the hiring of servants, and the payment of rents and other debts.

The English quarter days are:

Lady Day
: March 25, the Feast of the Annunciation.

Midsummer Day
: June 24, the Nativity of Saint John the Baptist.

Michaelmas
: September 29, the Feast of Saint Michael and All Angels.

Christmas
: December 25.

The traditional cross-quarter days fall approximately midway between the quarter days:

Candlemas
: February 2.

May Day
: May 1.

Lammas
: August 1.

All Hallows
: November 1.

The names and legal significance of these days vary among regions and historical periods.

6.2 Scottish Term and Quarter Days

The old Scottish term and quarter days are traditionally given as:

Candlemas
: February 2.

Whitsun
: May 15.

Lammas
: August 1.

Martinmas
: November 11.

Candlemas is associated with the Presentation of Christ and the Blessing of Candles. Whitsun derives from Pentecost, though the legal Scottish date became fixed rather than movable. Lammas takes its name from “loaf mass” and was associated with the first fruits of harvest. Martinmas is the feast of Saint Martin of Tours.

The Term and Quarter Days (Scotland) Act 1990 defines the modern Scottish quarter days as February 28, May 28, August 28, and November 28. May 28 and November 28 are the modern term days.

6.3 Uposatha

Uposatha is a recurring Buddhist day of observance. Monastics and lay followers may intensify meditation, study, generosity, and observance of the precepts.

The frequency and precise calculation of Uposatha vary among Buddhist traditions. In Theravada communities it is commonly associated with the new moon, full moon, and intervening quarter phases.

Four especially prominent full-moon observances are:

Magha Puja
: A commemoration of the early Sangha, ordinarily observed in February or March.

Vesak
: A principal commemoration of the Buddha, ordinarily observed in May.

Asalha Puja
: A commemoration of the first sermon, ordinarily observed in July.

Pavarana
: The observance marking the close of the rains retreat, ordinarily in October.

7 Alternative Calendars

7.1 Hanke-Henry Permanent Calendar

The Hanke-Henry Permanent Calendar is a proposed leap-week calendar. A common year contains 364 days, exactly 52 weeks. An additional week is inserted at intervals to keep the calendar aligned with the solar year.

Because every ordinary year has the same number of complete weeks, calendar dates fall on the same weekday from year to year. The broader proposal has also been associated with replacing civil time zones with Coordinated Universal Time.

7.2 Discordian Calendar

The Discordian or Erisian calendar is an alternative calendar described in the *Principia Discordia*. Its year begins on January 1 of the Gregorian calendar and contains five seasons of 73 days:

- Chaos;
- Discord;
- Confusion;
- Bureaucracy; and
- The Aftermath.

Discordian years are labelled YOLD, “Year of Our Lady of Discord.” The customary conversion adds 1166 to the Gregorian year.

The Discordian clock is a related alternative division of the day derived from the Erisian Klock specification.

8 Sources and Further Reading

The explanatory notes in the original `ind` man page were assembled chiefly from general reference material. This manual revises that prose and distinguishes technical definitions from historical summary.

Useful starting points include:

- United States Naval Observatory, *Introduction to Calendars*, `https://aa.usno.navy.mil/faq/calendars`.
- United States Naval Observatory, *Julian Date Converter*, `https://aa.usno.navy.mil/data/JulianDate`.
- United States Naval Observatory, *Astronomical Applications Glossary*, especially the entries for Julian Date and Modified Julian Date, `https://aa.usno.navy.mil/faq/asa_glossary`.
- United States Naval Observatory, *The Jewish Calendar*, `https://aa.usno.navy.mil/faq/jewish`.
- A. R. Chi, *A Grouped Binary Time Code for Telemetry and Space Application*, NASA, 1979.
- *Term and Quarter Days (Scotland) Act 1990*, 1990 c. 22.
- Edward M. Reingold and Nachum Dershowitz, *Calendrical Calculations*.

Wikipedia remains useful for orientation and for locating further sources, but it should not be treated as the sole authority for a technical or historical definition.

Concept Index

(Index is nonexistent)